

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

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NUMBER

PRELIMINARY EXAMINATION 2025 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

Paper 2

4045/02

5 August 2025

2 hours

Marking Scheme

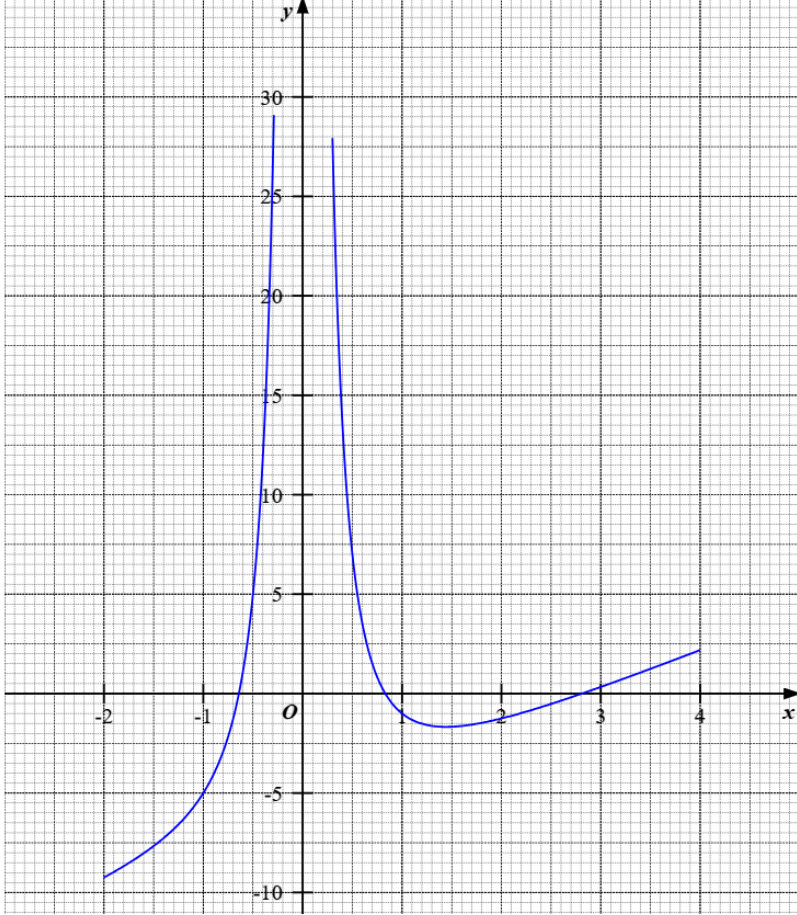
Qn	Answer	Marks
1a	3.01	1
1b	$\frac{\pi}{2}, \sqrt{2}, 120\%, -\left(\frac{5}{4}\right)^2$	2
2a	$\frac{a^8 \times a^2}{a^{-4}} = a^n$ $\frac{a^{10}}{a^{-4}} = a^n$	2
	$a^{10-(-4)} = a^n$ $n = 14$	

Qn	Answer	Marks
2b	$\sqrt{27} \div 9^2 = 3^m$ $3^{1.5} \div 3^4 = 3^m$	2
	$m = 1.5 - 4$ $= -2.5$	
3a	There are 15 students with at most 1 sibling and more than 18 students with more than or equal to 2 siblings. Middle position of data set will be greater than or equal to 2, no matter what value p takes.	1
3b	$\frac{9+26+3p+12+10}{6+9+13+p+3+2} = 1.95$ $\frac{57+3p}{33+p} = 1.95$ $57+3p = 64.35+1.95p$ $1.05p = 7.35$ $p = 7$	2
3c	$\frac{7+3+2}{6+9+13+7+3+2} = \frac{3}{10}$	1
Qn	Answer	Marks
4ai	Triangle B and triangle F	1
4aii	Triangle E	

4b	$\frac{PQ}{TS} = \frac{QR}{SQ}$ $\frac{8}{5} = \frac{QR}{12}$ $QR = 19.2 \text{ cm}$ $TR = 10.2 \text{ cm}$	3
5a	$\frac{5}{AC} = \sin 29^\circ$	2
	$AC = \frac{5}{\sin 29^\circ}$ $= 10.313 \text{ cm}$	

Qn	Answer	Marks
5b	$BC = \sqrt{10.313^2 - 5^2}$ $= 9.0198$	4
	Area of triangles $= 2 \times \frac{1}{2} \times 5 \times 9.0198$	
	Area of rectangles $= 13 \times (5 + 9.0198 + 10.313)$	
	Total surface area $= 361 \text{ cm}^2$ (3 s.f.)	
6a	$w - 4 = \frac{2w + 3}{x}$	3
	$x(w - 4) = 2w + 3$	
	$xw - 4x = 2w + 3$	
	$xw - 2w = 3 + 4x$	
	$w(x - 2) = 3 + 4x$	
	$w = \frac{3 + 4x}{x - 2}$	

Qn	Answer	Marks
6bi	$p(p - 6) + 12 = 2p$	2
	$p^2 - 6p + 12 = 2p$	
	$p^2 - 8p + 12 = 0$	
6bii	$(p - 6)(p - 2) = 0$	2
	$p = 6$ (rej.) or $p = 2$	
7a	2.2	1

Qn	Answer	Marks
7b		2
7c	When $x = 0$, $\frac{3}{x^2}$ is undefined. Hence, graph does not have a y-value when $x = 0$ and thus does not intersect y-axis.	1

Qn	Answer	Marks
7c	When $x = 0$, $\frac{3}{x^2}$ is undefined. Hence, graph does not have a y -value when $x = 0$ and thus does not intersect y -axis.	1
7d	balanced tangent drawn at $x = -1$	2
	Gradient = 8	
7e	$x = -0.65$ or 0.85 or 2.8	3

Qn	Answer	Marks
7c	When $x = 0$, $\frac{3}{x^2}$ is undefined. Hence, graph does not have a y -value when $x = 0$ and thus does not intersect y -axis.	1
7d	balanced tangent drawn at $x = -1$	2
	Gradient = 8	
7e	$x = -0.65$ or 0.85 or 2.8	3

Qn	Answer	Marks
8a	$\angle FAB = \frac{(6-2) \times 180^\circ}{6} = 120^\circ$	2
	$\angle ABF = \frac{180^\circ - 120^\circ}{2}$ (base angles of isos. triangle) $= 30^\circ$	
8b	$FE = BC$ (sides of a regular hexagon) $FB = EC$ (symmetry of regular hexagon) (or $FB = EC$ ($\triangle ABF$ congruent to $\triangle DEC$)) (or FB parallel to EC , FE parallel to BC)	2
	$\angle FBC = \angle ABC - 30^\circ$ $= 120^\circ - 30^\circ$ $= 90^\circ$	
	Since there are two pairs of equal sides and one of the interior angles of the rectangle is 90° , $BFEC$ is a rectangle.	
8c	Let midpoint of FB be M .	3
	$\frac{MB}{7} = \cos 30^\circ$	
	$MB = 6.0621 \text{ cm}$ $FB = 12.124 \text{ cm}$	
	Area $= 12.124 \times 7$ $= 84.9 \text{ cm}^2$ (3 s.f.)	

Qn	Answer	Marks
9a	$AB = \sqrt{(5-1)^2 + [1-(-1)]^2}$	2
	$= 4.47 \text{ units}$	
9b	gradient $= \frac{-1-(-3)}{1-5}$	3
	$= -0.5$	
	$y = -0.5x + c$ Sub $(1, -1)$: $-1 = -0.5 + c$ $c = -0.5$	
	equation: $y = -0.5x - 0.5$	
9ci	$(11, -1)$	1
Qn	Answer	Marks
9cii	$y = -1$	1
9ciii	Area of $\triangle ABD = \frac{1}{2} \times 10 \times 2 = 10 \text{ units}^2$	2
	Area of kite $= 20 \text{ units}^2$	

Qn	Answer	Marks
10a	$\frac{18}{100} \times 275.1 = 49.518 \text{ kWh}$	1
10b	$29.67 \times 379 \times 12 \times \frac{109}{100} \div 100$	3
	= \$1471 (nearest dollar)	
10ci	Estimated aircon usage per month $= \frac{36}{100} \times 500 = 180 \text{ kWh}$	1
10cii	Suggested aircon usage per month $= \frac{5}{8} \times 180 = 112.5 \text{ kWh}$	3
	Total usage after suggestion $= (500 - 180) + 112.5$ $= 432.5 \text{ kWh}$	
	Since $432.5 < 461.4$, Mr Tan is correct.	

Section B

Qn	Answer	Marks
11ai	155 g	1
11aii	$212 \text{ g} - 110 \text{ g} = 102 \text{ g}$	1
11aiii	$170 \text{ g} - 126 \text{ g} = 44 \text{ g}$	1
11aiv	Her statement is incorrect as the interquartile range of apple masses from farm B is greater at 50 g than that of farm A at 44 g	1
11bi	The data is grouped and thus the exact height of trees is not known. <u>The shortest tree could be greater than 1 metre in height and the tallest tree could be shorter than 6 metres in height, which will give a range of heights smaller than 5 metres.</u>	1
11bii	$2 + 5 + 11 = 18$ $\frac{18}{50} \times \frac{13}{49} + \frac{13}{50} \times \frac{18}{49}$	3
	$= \frac{234}{1225}$	

Qn	Answer	Marks
12ai	$\angle RPQ = 0.4$ radians	1
	Reason angles in the same segment are equal	
12aii	$\angle SPR = \frac{\pi}{2} - 1.1$ (tangent perpendicular to radius) $= 0.471$	1
12bi	$7.8^2 = 6.8^2 + 10.8^2 - 2(6.8)(10.8)\cos \angle BAC$	4
	$\cos \angle BAC = \frac{6.8^2 + 10.8^2 - 7.8^2}{2(6.8)(10.8)}$	
	$\angle BAC = \cos^{-1}\left(\frac{102.04}{146.88}\right)$ $= 45.995^\circ$	
	Bearing of B from $A = 90^\circ - 45.995^\circ$	
	$= 044.0^\circ$ (1 d.p.)	
12bii	Let angle of elevation be x .	2
	$\tan x = \frac{4.6}{7.8}$	
	$x = \tan^{-1}\left(\frac{4.6}{7.8}\right)$ $= 30.5^\circ$ (1 d.p.)	
– 1m from entire question if no degree symbol written at all – 1m from entire question if answers of angles not in 1 d.p.		